

1. Find the value of the following limit: $\lim_{x \rightarrow 5} (2^{(9-x)})$.
2. A sequence is defined recursively by $a_{(n+1)} = 3a_n - 5$ for all integers $n \geq 1$. If $a_{10} = 16$, find the value of a_8 .
3. Let $\vec{a}$ and $\vec{b}$ represent vectors such that $\vec{a} = (3, 2)$ and $\vec{b} = (-8, 13)$. Find the **ordered pair** that represents $3\vec{a} + 5\vec{b}$.
4. Find the value of the indicated sum: $\sum_2^5 (k^3 + 3.782)$. Express your answer as an **exact decimal**.
5. Find the tenth term of the geometric sequence whose first three terms are respectively: $\frac{1024}{5}$, $\frac{512}{15}$, and $\frac{256}{45}$. Express your answer as a common fraction reduced to lowest terms.
6. Given the relation defined by $f = \{(2, 7), (5, 3), (7, 11), (19, w)\}$. Find the product of all the distinct real values of w for which f^{-1} is **not** a function.
7. Let k be a positive integer and let w be an integer such that two of the roots for x of the cubic equation $x^3 - 17x^2 + kx + w = 0$ are consecutive positive integers. Find the value of k if $|k - w|$ is a maximum.